

TABLE 2.3 Summary of Important Parameters and Associated Formulas and Equation Numbers

Parameter	Formula	Equation Number
Infinitesimal area of sphere	$dA = r^2 \sin \theta d\theta d\phi$	(2-1)
Elemental solid angle of sphere	$d\Omega = \sin \theta d\theta d\phi$	(2-2)
Average power density	$\mathbf{W}_{av} = \frac{1}{2} \text{Re}[\mathbf{E} \times \mathbf{H}^*]$	(2-8)
Radiated power/average radiated power	$P_{rad} = P_{av} = \iint_S \mathbf{W}_{av} \cdot d\mathbf{s} = \frac{1}{2} \iint_S \text{Re}[\mathbf{E} \times \mathbf{H}^*] \cdot d\mathbf{s}$	(2-9)
Radiation density of isotropic radiator	$W_0 = \frac{P_{rad}}{4\pi r^2}$	(2-11)
Radiation intensity (far field)	$U = r^2 W_{rad} = B_0 F(\theta, \phi) \simeq \frac{r^2}{2\eta}$	(2-12),
	$\times [E_\theta(r, \theta, \phi) ^2 + E_\phi(r, \theta, \phi) ^2]$	(2-12a)
Directivity $D(\theta, \phi)$	$D = \frac{U}{U_0} = \frac{4\pi U}{P_{rad}} = \frac{4\pi}{\Omega_A}$	(2-16),
		(2-23)
Beam solid angle Ω_A	$\Omega_A = \int_0^{2\pi} \int_0^\pi F_n(\theta, \phi) \sin \theta d\theta d\phi$	(2-24)
	$F_n(\theta, \phi) = \frac{F(\theta, \phi)}{ F(\theta, \phi) _{\max}}$	(2-25)
Maximum directivity D_0	$D_{\max} = D_0 = \frac{U_{\max}}{U_0} = \frac{4\pi U_{\max}}{P_{rad}}$	(2-16a)
Partial directivities D_θ, D_ϕ	$D_0 = D_\theta + D_\phi$	(2-17)
	$D_\theta = \frac{4\pi U_\theta}{P_{rad}} = \frac{4\pi U_\theta}{(P_{rad})_\theta + (P_{rad})_\phi}$	(2-17a)
	$D_\phi = \frac{4\pi U_\phi}{P_{rad}} = \frac{4\pi U_\phi}{(P_{rad})_\theta + (P_{rad})_\phi}$	(2-17b)
Approximate maximum directivity (one main lobe pattern)	$D_0 \simeq \frac{4\pi}{\Theta_{1r}\Theta_{2r}} = \frac{41,253}{\Theta_{1d}\Theta_{2d}}$	(2-26),
	(Kraus)	(2-27)
	$D_0 \simeq \frac{32 \ln 2}{\Theta_{1r}^2 + \Theta_{2r}^2} = \frac{22.181}{\Theta_{1r}^2 + \Theta_{2r}^2} = \frac{72,815}{\Theta_{1d}^2 + \Theta_{2d}^2}$	(2-30),
	(Tai-Pereira)	(2-30a),
		(2-30b)
Approximate maximum directivity (omnidirectional pattern)	$D_0 \simeq \frac{101}{\text{HPBW}(\text{degrees}) - 0.0027[\text{HPBW}(\text{degrees})]^2}$	(2-33a)
	(McDonald)	
	$D_0 \simeq -172.4 + 191 \sqrt{0.818 + \frac{1}{\text{HPBW}(\text{degrees})}}$	(2-33b)
	(Poazar)	

(continued overleaf)

TABLE 2.3 (continued)

Parameter	Formula	Equation Number
Gain $G(\theta, \phi)$	$G = \frac{4\pi U(\theta, \phi)}{P_{in}} = e_{cd} \left[\frac{4\pi U(\theta, \phi)}{P_{rad}} \right] = e_{cd} D(\theta, \phi)$	(2-46),
		(2-47)
	$P_{rad} = e_{cd} P_{in}$	(2-49)
Antenna radiation efficiency e_{cd}	$e_{cd} = \frac{R_r}{R_r + R_L}$	(2-90)
Loss resistance R_L (straight wire/uniform current)	$R_L = R_{hf} = \frac{l}{P} \sqrt{\frac{\omega \mu_0}{2\sigma}}$	(2-90b)
Loss resistance R_L (straight wire/ $\lambda/2$ dipole)	$R_L = \frac{l}{2P} \sqrt{\frac{\omega \mu_0}{2\sigma}}$	
Maximum gain G_0	$G_0 = e_{cd} D_{max} = e_{cd} D_0$	(2-49a)
Partial gains G_θ, G_ϕ	$G_0 = G_\theta + G_\phi$	(2-50)
		(2-50a),
	$G_\theta = \frac{4\pi U_\theta}{P_{in}}, \quad G_\phi = \frac{4\pi U_\phi}{P_{in}}$	(2-50b)
Realized gain G_{re}	$G_{re} = e_r G(\theta, \phi) = e_r e_{cd} D(\theta, \phi) = (1 - \Gamma ^2) e_{cd} D(\theta, \phi)$	(2-49a)
	$= e_0 D(\theta, \phi)$	(2-49b)
Total antenna efficiency e_0	$e_0 = e_r e_c e_d = e_r e_{cd} = (1 - \Gamma ^2) e_{cd}$	(2-52)
Reflection efficiency e_r	$e_r = (1 - \Gamma ^2)$	(2-45)
Beam efficiency BE	$BE = \frac{\int_0^{2\pi} \int_0^{\theta_1} U(\theta, \phi) \sin \theta d\theta d\phi}{\int_0^{2\pi} \int_0^\pi U(\theta, \phi) \sin \theta d\theta d\phi}$	(2-54)
Polarization loss factor (PLF)	$PLF = \hat{\rho}_w \cdot \hat{\rho}_a ^2$	(2-71)
Vector effective length $\mathcal{L}_e(\theta, \phi)$	$\mathcal{L}_e(\theta, \phi) = \hat{\mathbf{a}}_\theta l_\theta(\theta, \phi) + \hat{\mathbf{a}}_\phi l_\phi(\theta, \phi)$	(2-91)
Polarization efficiency p_e	$p_e = \frac{ \mathcal{L}_e \cdot \mathbf{E}^{inc} ^2}{ \mathcal{L}_e ^2 \mathbf{E}^{inc} ^2}$	(2-71a)
Antenna impedance Z_A	$Z_A = R_A + jX_A = (R_r + R_L) + jX_A$	(2-72),
		(2-73)
Maximum effective area A_{em}	$A_{em} = \frac{ V_T ^2}{8W_i} \left[\frac{1}{R_r + R_L} \right] = e_{cd} \left(\frac{\lambda^2}{4\pi} \right) D_0 \hat{\rho}_w \cdot \hat{\rho}_a ^2$	(2-96),
		(2-111),
	$= \left(\frac{\lambda^2}{4\pi} \right) G_0 \hat{\rho}_w \cdot \hat{\rho}_a ^2$	(2-112)
Aperture efficiency ϵ_{ap}	$\epsilon_{ap} = \frac{A_{em}}{A_p} = \frac{\text{maximum effective area}}{\text{physical area}}$	(2-100)
Friis transmission equation	$\frac{P_r}{P_t} = \left(\frac{\lambda}{4\pi R} \right)^2 G_{0t} G_{0r} \hat{\rho}_t \cdot \hat{\rho}_r ^2$	(2-118),
		(2-119)
Radar range equation	$\frac{P_r}{P_t} = \sigma \frac{G_{0t} G_{0r}}{4\pi} \left[\frac{\lambda}{4\pi R_1 R_2} \right]^2 \hat{\rho}_w \cdot \hat{\rho}_r ^2$	(2-125),
		(2-126)